Analysis of OMICS data

Lecture #11
single cell transcriptomics

from individual cells to data
from data to knowledge

15may-2026

ELTE EÖTVÖS LORÁND UNIVERSITY

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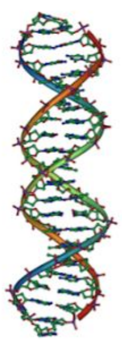
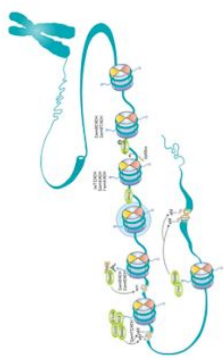
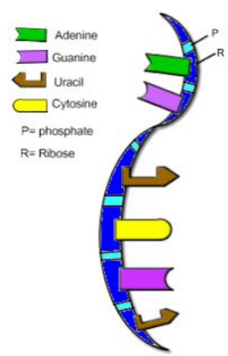
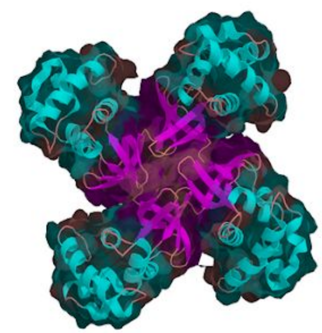
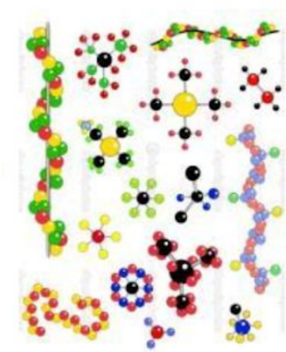
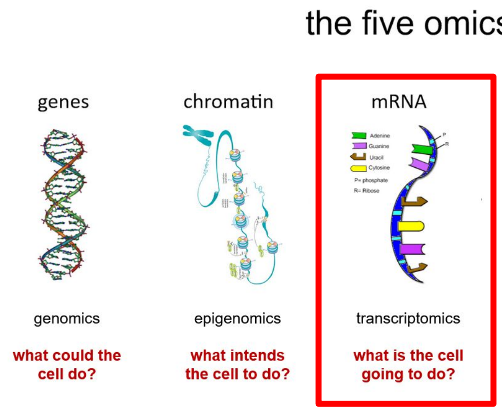
~~the Central Dogma of Molecular Biology~~

the five omics

genes	chromatin	mRNA	proteins	metabolites
				
genomics	epigenomics	transcriptomics	proteomics	metabolomics
what could the cell do?	**what intends the cell to do?**	**what is the cell going to do?**	**what is the cell *actually* doing?**	**what did the cell do?**

Why the mess of working with single cells?

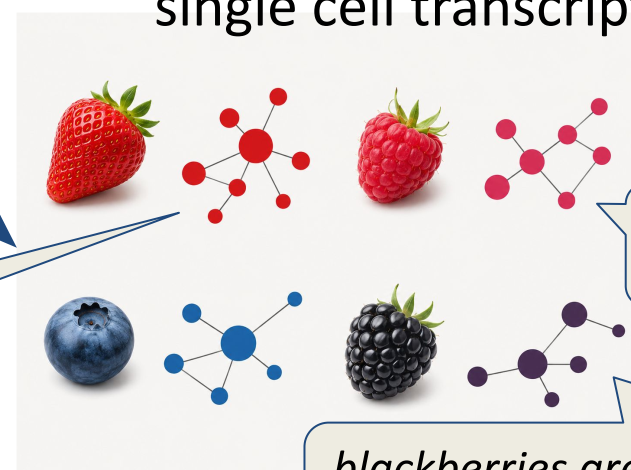


the smoothie is sweeter

bulk transcriptomics

organ/tissue

single cell transcriptomics

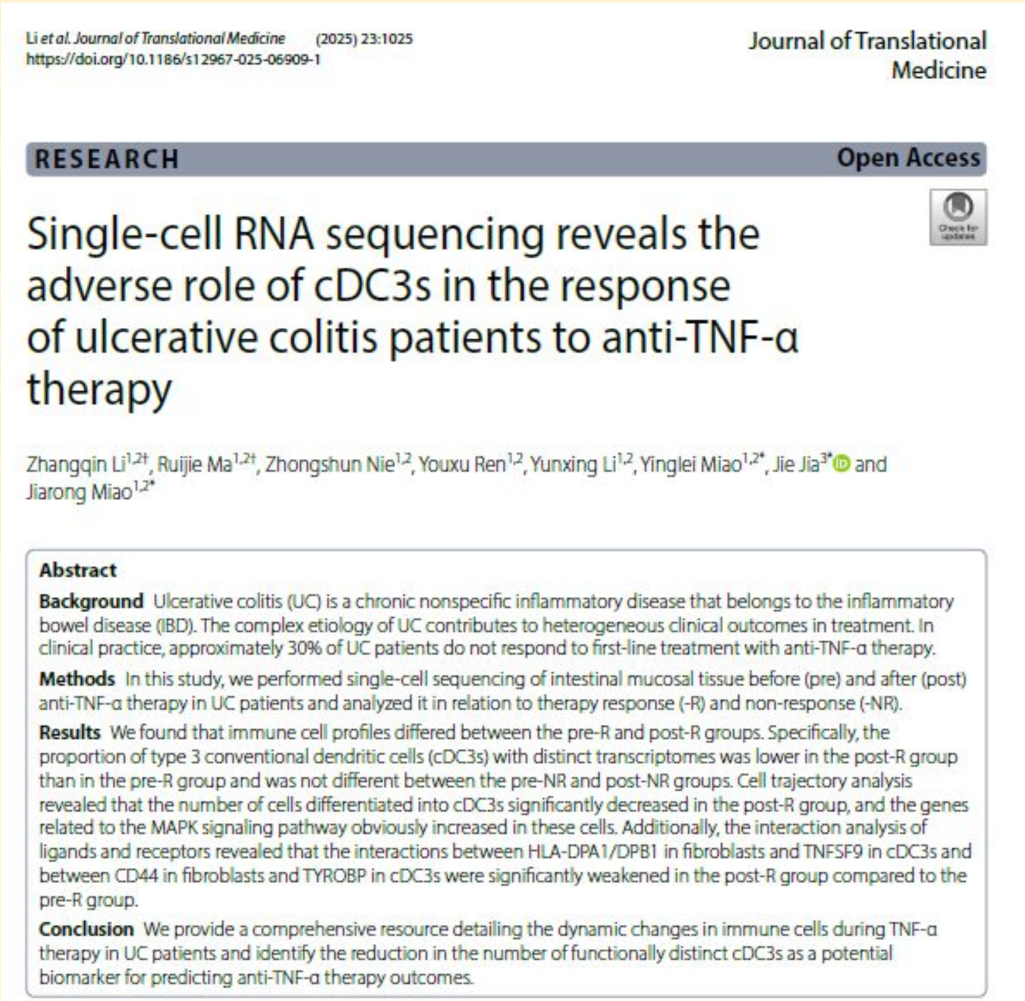


raspberries are just fine

strawberries are sweeter, because they are ripe

blackberries are sour, but they are just a few

Some single cell transcriptomics wins


Li et al. Journal of Translational Medicine (2025) 23:1025
https://doi.org/10.1186/s12967-025-06909-1

Journal of Translational Medicine

RESEARCH Open Access

Single-cell RNA sequencing reveals the adverse role of cDC3s in the response of ulcerative colitis patients to anti-TNF-α therapy

Zhangqin Li[1,2†], Ruijie Ma[1,2†], Zhongshun Nie[1,2], Youxu Ren[1,2], Yunxing Li[1,2], Yinglei Miao[1,2*], Jie Jia[3*] and Jiarong Miao[1,2*]

Abstract

Background Ulcerative colitis (UC) is a chronic nonspecific inflammatory disease that belongs to the inflammatory bowel disease (IBD). The complex etiology of UC contributes to heterogeneous clinical outcomes in treatment. In clinical practice, approximately 30% of UC patients do not respond to first-line treatment with anti-TNF-α therapy.

Methods In this study, we performed single-cell sequencing of intestinal mucosal tissue before (pre) and after (post) anti-TNF-α therapy in UC patients and analyzed it in relation to therapy response (-R) and non-response (-NR).

Results We found that immune cell profiles differed between the pre-R and post-R groups. Specifically, the proportion of type 3 conventional dendritic cells (cDC3s) with distinct transcriptomes was lower in the post-R group than in the pre-R group and was not different between the pre-NR and post-NR groups. Cell trajectory analysis revealed that the number of cells differentiated into cDC3s significantly decreased in the post-R group, and the genes related to the MAPK signaling pathway obviously increased in these cells. Additionally, the interaction analysis of ligands and receptors revealed that the interactions between HLA-DPA1/DPB1 in fibroblasts and TNFSF9 in cDC3s and between CD44 in fibroblasts and TYROBP in cDC3s were significantly weakened in the post-R group compared to the pre-R group.

Conclusion We provide a comprehensive resource detailing the dynamic changes in immune cells during TNF-α therapy in UC patients and identify the reduction in the number of functionally distinct cDC3s as a potential biomarker for predicting anti-TNF-α therapy outcomes.

https://link.springer.com/article/10.1186/s12967-025-06909-1 -> «Single-cell RNA sequencing reveals the adverse role of cDC3s in the response of ulcerative colitis patients to anti-TNF-α therapy» (2025)

#1 scRNA-seq revealed that patients with sustained tissue inflammation after first-line biologic therapy showed persistent inflammatory dendritic cells.

Some single cell transcriptomics wins

#2 blood scRNA-seq during induction therapy showed persistent antiviral monocyte programs mark early resistance in lupus nephritis.

Ann Rheum Dis 85 (2026) 647–658

Contents lists available at ScienceDirect

Annals of the Rheumatic Diseases

journal homepage: https://www.sciencedirect.com/journal/annals-of-the-rheumatic-diseases

ELSEVIER

Systemic lupus erythematosus

Treatment nonresponders in lupus nephritis characteristically exhibit persistent type I interferon signalling in monocytes: a longitudinal single-cell transcriptomic analysis

Woo-Joong Kim[1,2,†], Hye-Soon Lee[3,4,5,†], So-Young Bang[3,4,5], Young Bin Joo[3,4,5], Bo-Kyeong Kang[6], Mimi Kim[6], Hyunsung Kim[7], Sung Yul Park[8], Woong-Yang Park[9], Eui-Cheol Shin[1,‡,**], Sang-Cheol Bae[3,4,5,‡,*]

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https://www.sciencedirect.com/science/article/abs/pii/S0003496725045960 -> «Treatment nonresponders in lupus nephritis characteristically exhibit persistent type I interferon signalling in monocytes: a longitudinal single-cell transcriptomic analysis» (2026)

Some single cell transcriptomics wins

nature medicine

Article https://doi.org/10.1038/s41591-025-03574-1

Microglial mechanisms drive amyloid-β clearance in immunized patients with Alzheimer's disease

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Check for updates

A list of authors and their affiliations appears at the end of the paper

Alzheimer's disease (AD) therapies utilizing amyloid-β (Aβ) immunization have shown potential in clinical trials. Yet, the mechanisms driving Aβ clearance in the immunized AD brain remain unclear. Here, we use spatial transcriptomics to explore the effects of both active and passive Aβ immunization in the AD brain. We compare actively immunized patients with AD with nonimmunized patients with AD and neurologically healthy controls, identifying distinct microglial states associated with Aβ clearance. Using high-resolution spatial transcriptomics alongside single-cell RNA sequencing, we delve deeper into the transcriptional pathways involved in Aβ removal after lecanemab treatment. We uncover spatially distinct microglial responses that vary by brain region. Our analysis reveals upregulation of the triggering receptor expressed on myeloid cells 2 (*TREM2*) and apolipoprotein E (*APOE*) in microglia across immunization approaches, which correlate positively with antibody responses and Aβ removal. Furthermore, we show that complement signaling in brain myeloid cells contributes to Aβ clearance after immunization. These findings provide new insights into the transcriptional mechanisms orchestrating Aβ removal and shed light on the role of microglia in immune-mediated Aβ

https://www.nature.com/articles/s41591-025-03574-1 -> «Microglial mechanisms drive amyloid-β clearance in immunized patients with Alzheimer's disease» (2025)

#3 spatial single-cell profiling identified region-specific brain immune-cell states linked to amyloid clearance after Alzheimer immunotherapy.

Some single cell transcriptomics wins

#4 pregnancy is… complicated: a signalling network between two organisms. Single cell RNA-seq was essential because it showed evolution developed an asymmetric relationship.

nature ecology & evolution

Article

https://doi.org/10.1038/s41559-025-02748-x

Cell type and cell signalling innovations underlying mammalian pregnancy

Received: 11 September 2024

Accepted: 12 May 2025

Published online: 1 July 2025

Check for updates

Daniel J. Stadtmauer[1,2,3,8], Silvia Basanta[3,8], Jamie D. Maziarz[1], Alison G. Cole[4], Gülay Dagdas[3], Gilbecca Rae Smith[5], Frank van Breukelen[5], Mihaela Pavličev[2,3,6] & Günter P. Wagner[1,2,3,7]

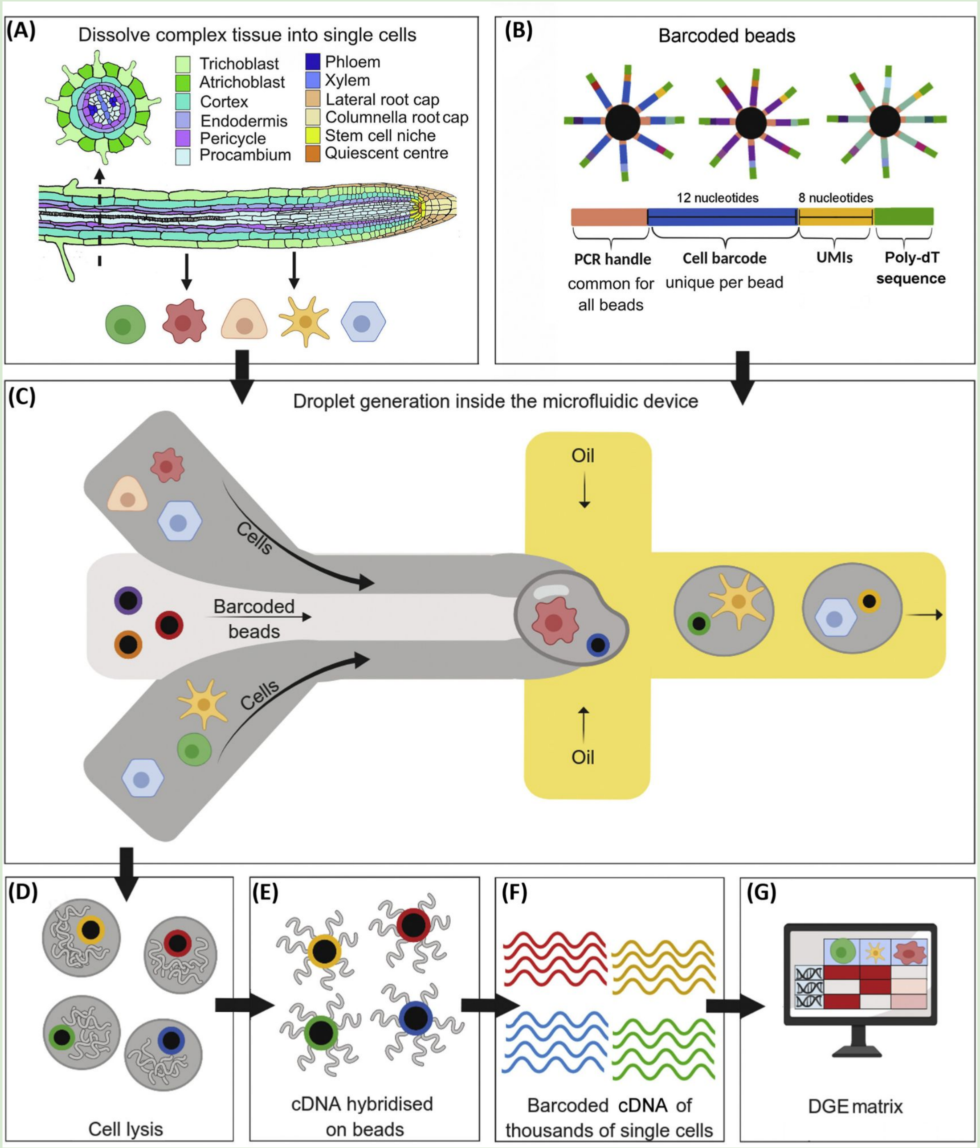
How fetal and maternal cell types have co-evolved to enable mammalian placentation poses a unique evolutionary puzzle. Here we integrate and compare single-cell transcriptomes from six species bracketing therian mammal diversity: opossum (a marsupial), Malagasy common tenrec (an afrotherian), mouse and guinea pig (rodents), and macaque and human (primates). We identify a conserved transcriptomic signature of invasive trophoblast across eutherians, probably representing a cell type family that radiated with the evolution of haemochorial placentation. In the maternal stroma, comparative analysis reveals that the endocrine decidual cell evolved from an immunomodulatory predecidual cell type retained in *Tenrec* and resembling early human decidua. Fetal and maternal cell signalling shows a pronounced tendency towards disambiguation—the exclusive expression of ligands by only one partner—although few ligand–receptor pairs follow an escalatory arms race dynamic. Finally, we reconstruct the uteroplacental cell–cell communication networks of extinct mammalian ancestors, identifying signalling innovations and widespread integration of fetal trophoblast and maternal decidual cells into signalling networks. Together, these results reveal a dynamic history of cell type innovation and co-evolution at the fetal–maternal interface.

https://www.nature.com/articles/s41559-025-02748-x -> «Cell type and cell signalling innovations underlying mammalian pregnancy» (2025)

Single cell transcriptomics overview

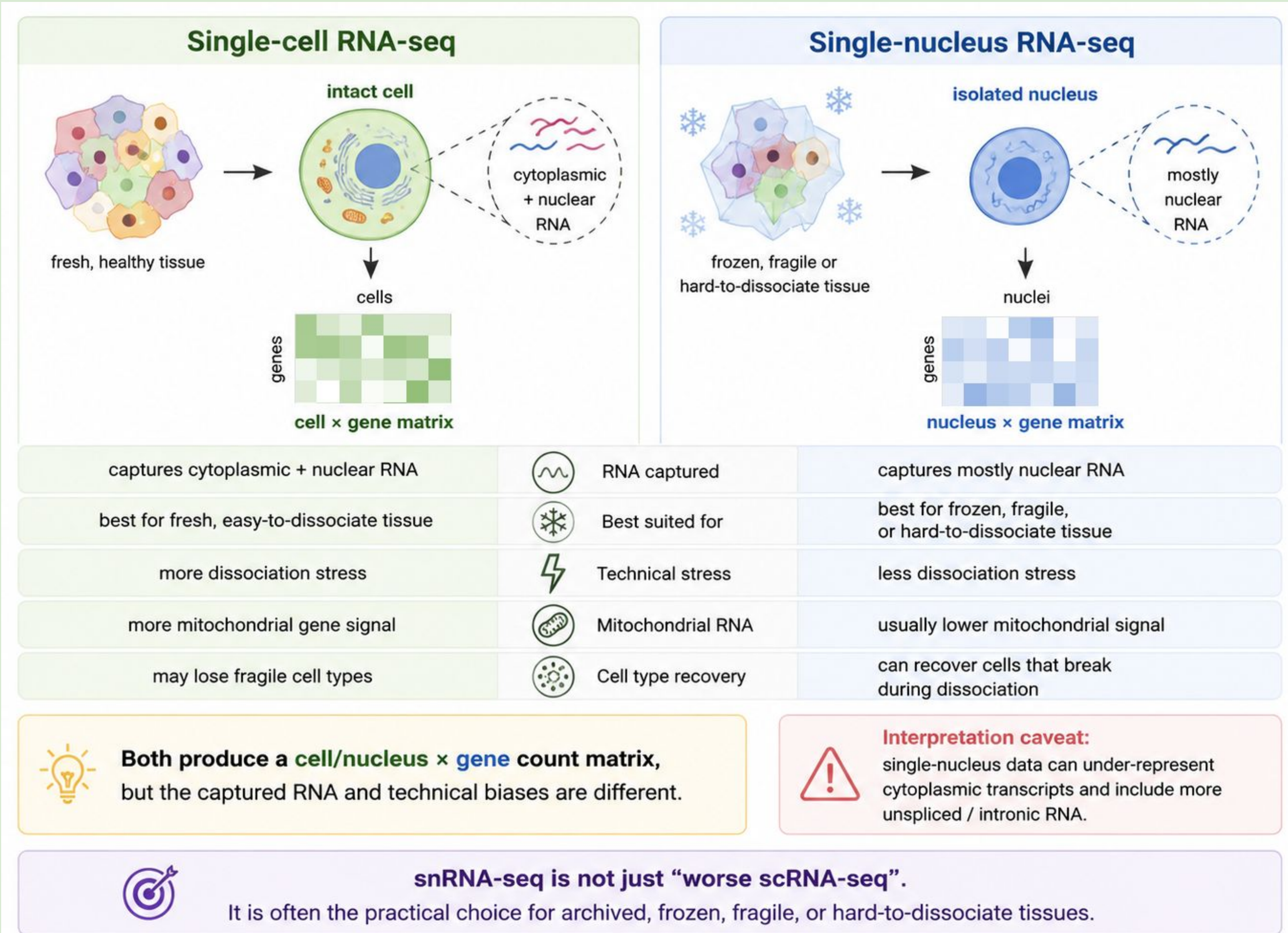
From tissue to droplets

From droplets to data files



(Rich-Griffin 2020)

Single cell vs single nucleus

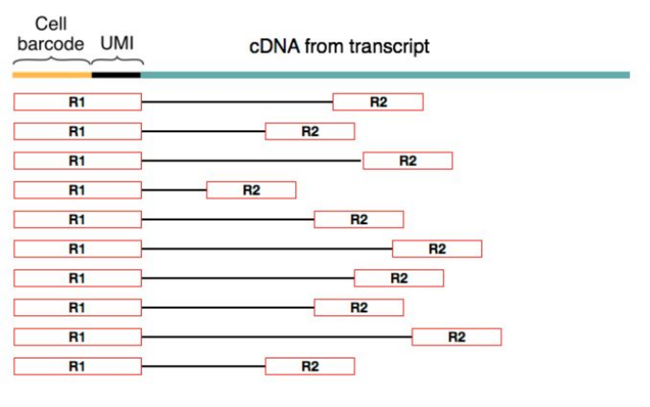


Barcodes and UMIs

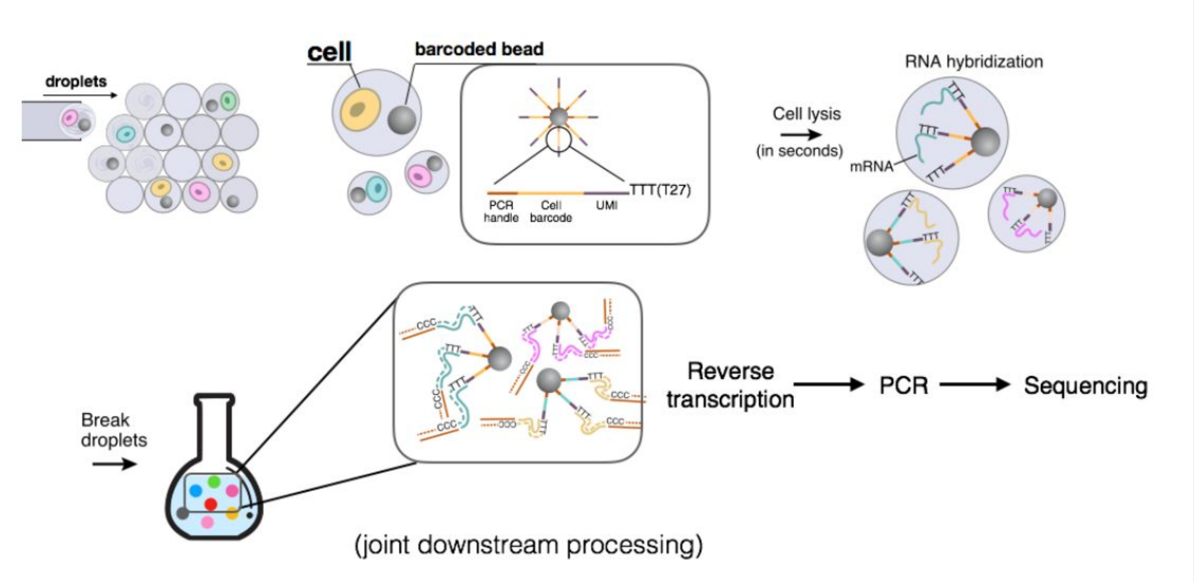
Each droplet ideally contains one cell/nucleus and one bead.
Each bead carries many oligo-dT primers.
All primers on the same bead share the same **cell barcode** + **different UMI each** + **a poly(dT) sequence** that hybridises to the poly(A) tail of a mRNA molecule.

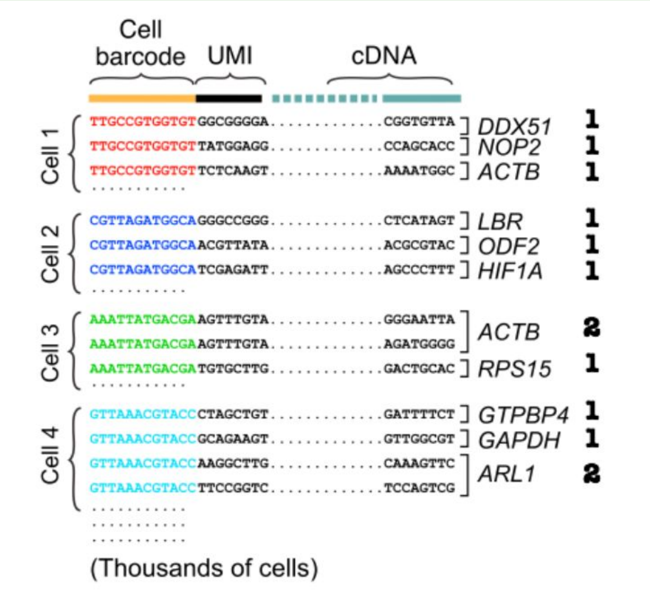
Cell barcode: labels the cell/nucleus.
UMI (unique molecule identifier): labels an mRNA molecule.

Together > **cell × gene count matrix**.



Biased paired-end reads.





Source of images:
https://data-science-sequencing.github.io/Win2018/lectures/lecture16/

From sequencing reads to matrices

How are the starting datasets processed before we start working on them?

sc RNA-seq or sn RNA-seq experiment	*sequencing and creation of FASTQ files*	*read mapping with Cell Ranger, STARsolo, or another aligner*	*alignments / BAM files* *cell/nucleus calling from raw droplet data*	*creation of count matrices with cells × genes or nuclei × genes*

We will start here, but there is a lot of work before that!

cell × gene matrices

in our matrix this will be our ".X" table

raw expression counts = how many transcripts were assigned to each gene in each cell/nucleus

each one is a cell or a nucleus

GENES

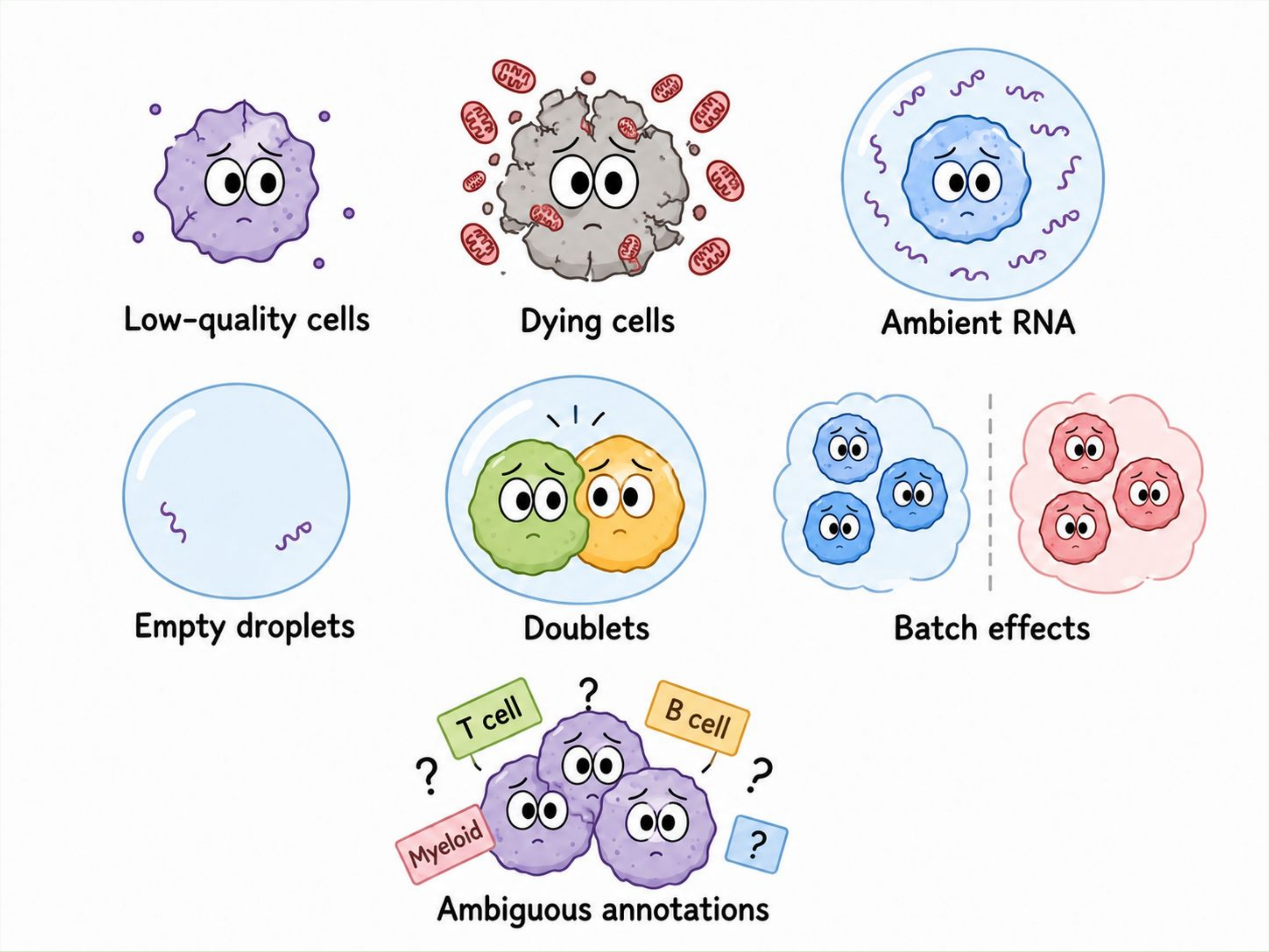
BARCODES	MSTO1	YY1AP1	DAP3	AL162734.1	GON4L	SYT11	RIT1	KHDC4	RXFP4	ARHGEF2	SSR2	UBQLN4	LAMTOR2		MEX3A	LMNA	SEMA4A	SLC25A44
CCTTTGGAGTGCCAGA	0	0	0	0	0	0	3191	0	0	0	8980	0	0	[illegible]	0	0	0	0
CCTTTGGCAGTCCGTG	0	1	0	0	0	0	0	0	0	0	7906	0	0	0	0	0	0	0
CCTTTGGCATACAGCT	0	0	0	0	1	0	3683	0	1	0	0	0	0	0	0	0	0	0
CCTTTGGCATGGCGCT	0	0	0	0	0	0	4144	0	0	0	9558	0	0	0	0	0	0	0
CCTTTGGGTATTAAGG	0	0	0	3	0	0	4239	0	0	0	7410	0	0	0	0	0	0	1
CCTTTGGGTCGCCACA	0	0	0	0	0	1	0	0	0	0	0	3	0	0	0	0	0	0
CCTTTGGGTGAGCTCC	0	0	0	0	0	0	3734	0	0	0	8376	0	0	0	0	1	0	0
CGAAGGAAGACGAAGA	1	0	0	0	0	0	3819	0	0	0	2	0	0	2	0	0	0	0
CGAAGGAAGCACCTGC	0	0	0	0	0	4	2965	0	0	0	9804	0	0	0	0	0	0	0
CGAAGGAAGCACTCAT	0	0	0	0	0	0	0	0	0	0	7803	0	0	0	0	0	0	0
CGAAGGAAGGTAAGTT	0	0	0	0	0	0	2060	0	0	0	0	0	0	0	0	0	4	0
CGAAGGACATCGTGCG	0	0	2	0	0	0	2	0	0	0	6370	0	0	0	0	0	0	0
CGAAGGAGTAAGGAGA	0	0	0	0	0	0	478	0	0	0	9446	0	0	0	0	0	0	0
CGAAGGAGTACCGGCT	1	0	0	0	0	3	0	0	0	0	7899	0	0	0	0	0	0	0
CGAAGGAGTCAGTCCG	0	0	0	0	1	2	1104	0	0	0	2	0	0	0	0	0	0	0
CGAAGGATCCGATCGG	0	0	0	0	0	0	595	0	0	1	7080	[illegible]	4	2	1	0	0	1
CGAAGGATCGTTCCTG	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CGAAGGATCTCTTGCG	0	0	0	0	0	0	1893	0	0	0	6617	0	[illegible]	0	0	0	0	0
CGAAGTTCACTTGGGC	0	0	0	0	0	0	4299	2	0	0	1	0	0	[illegible]	0	0	0	0
CGAAGTTCATATGGC	0	0	[illegible]	0	0	0	1767	0	2	0	8834	0	0	[illegible]	0	0	0	0

in our matrix, we will have also some metadata for each cell/nucleus

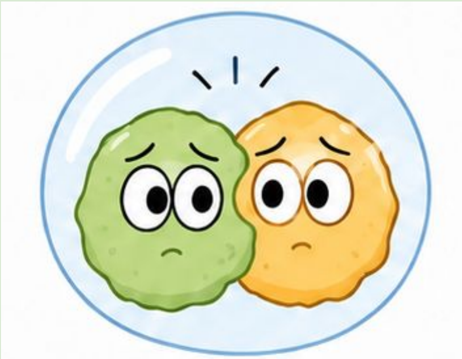
note that it's a sparse matrix: most values are just zero

we will disentangle the biology that shows up when a gene is found abundantly in many cells

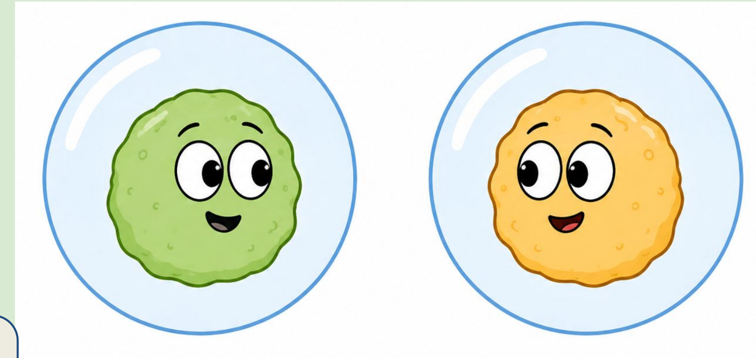
When things go wrong…



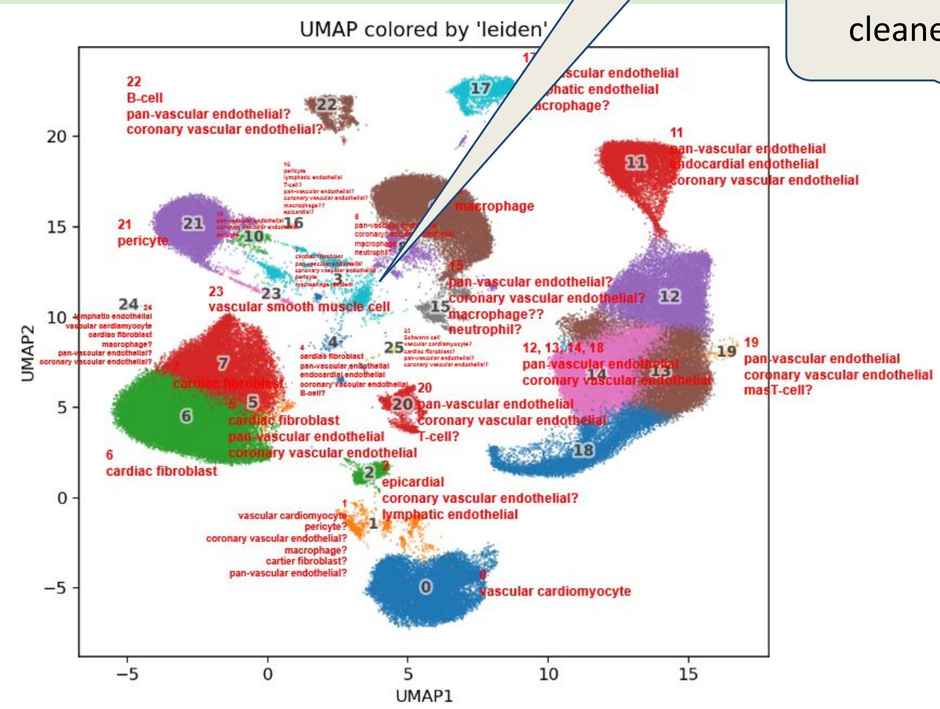
The problem of doublets

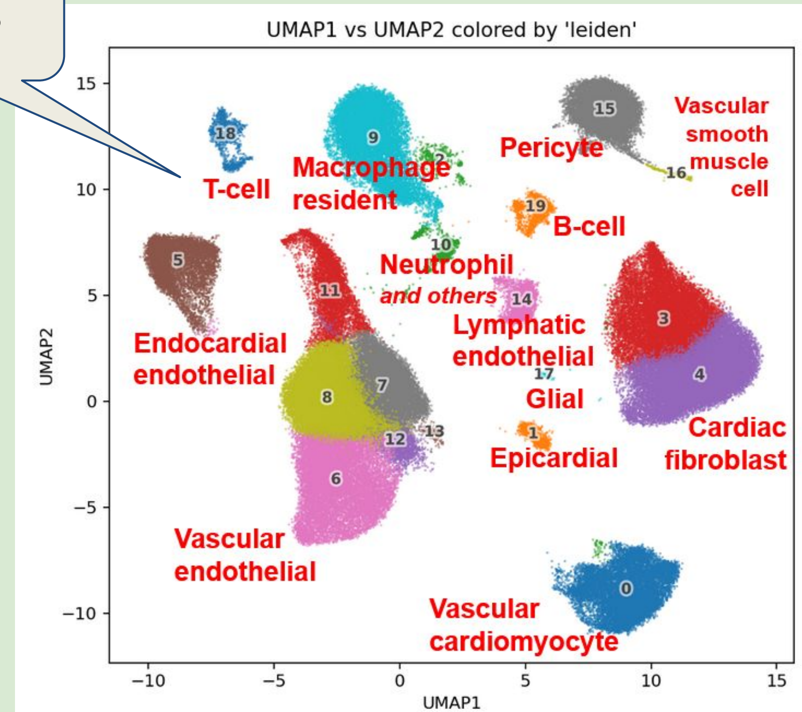


such a mess when I forgot to remove the doublets!

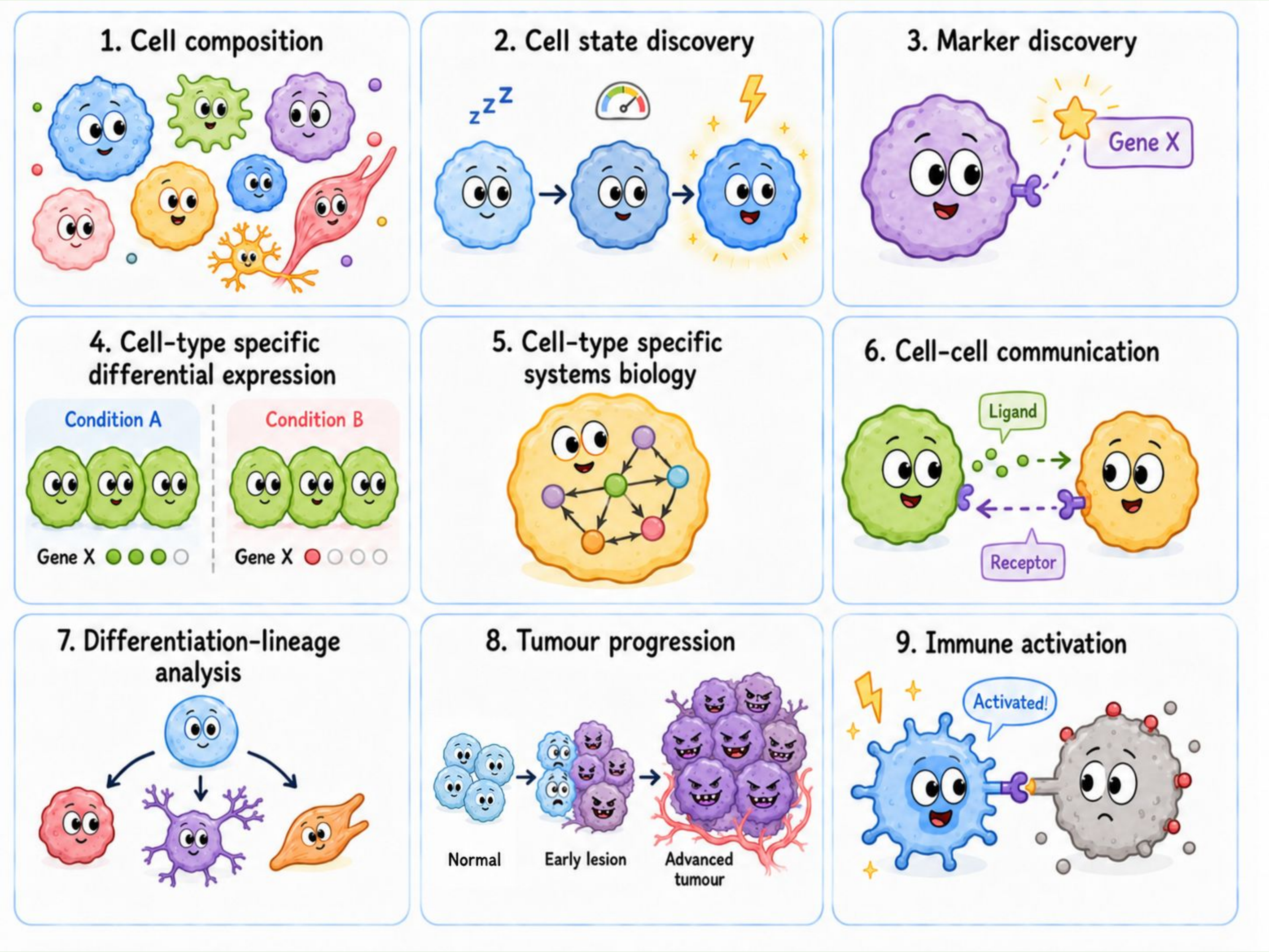


isn’t it much cleaner?





Why do we do all this?



Why do we still use bulk RNAseq?

	Bulk RNA-seq	Single cell RNA-seq
heterogeneity	everything is mixed into the average, so the most interesting effects might be lost	can separate cell composition or cell state changes
price	cheaper	more expensive
experimental complexity	simpler	more difficult
replicates	you can afford more replicates	less replicates
resolution	can detect low expressed genes	only highly expressed genes
statistics	more statistical power	less statistical power
interpretation	easier: one expression profile per sample	harder but richer: you can find the expression of each cell type, but you can have strong batch effects
focus	tissue level: for many experiments you don't need single cell	cell level: only if you need to resolve the individual cells
sample constraints	you can almost always get everything from the sample	some samples only allow single nucleus, not whole cell
screening	useful to select the best samples to do later single cell RNA-seq	better after you already know which are he most promising samples

Computational workflow overview

sc RNA-seq or sn RNA-seq experiment

sequencing and creation of FASTQ files

read mapping with Cell Ranger or another aligner

alignments / BAM files

cell/nucleus calling from raw droplet data

creation of count matrices with cells × genes or nuclei × genes

Task 1
just merging files
scanpy
pandas

Task 2
quality control

Task 3
cell demographics
matplotlib

Task 4
pseudobulk RNA-seq
numpy
DESeq2
(via pydeseq2)

Task 5
doublet tagging and removal
Scrublet

Task 6
creating a UMAP (and failing)

Task 7
correction of batch-effects
Harmony
(via harmonypy)

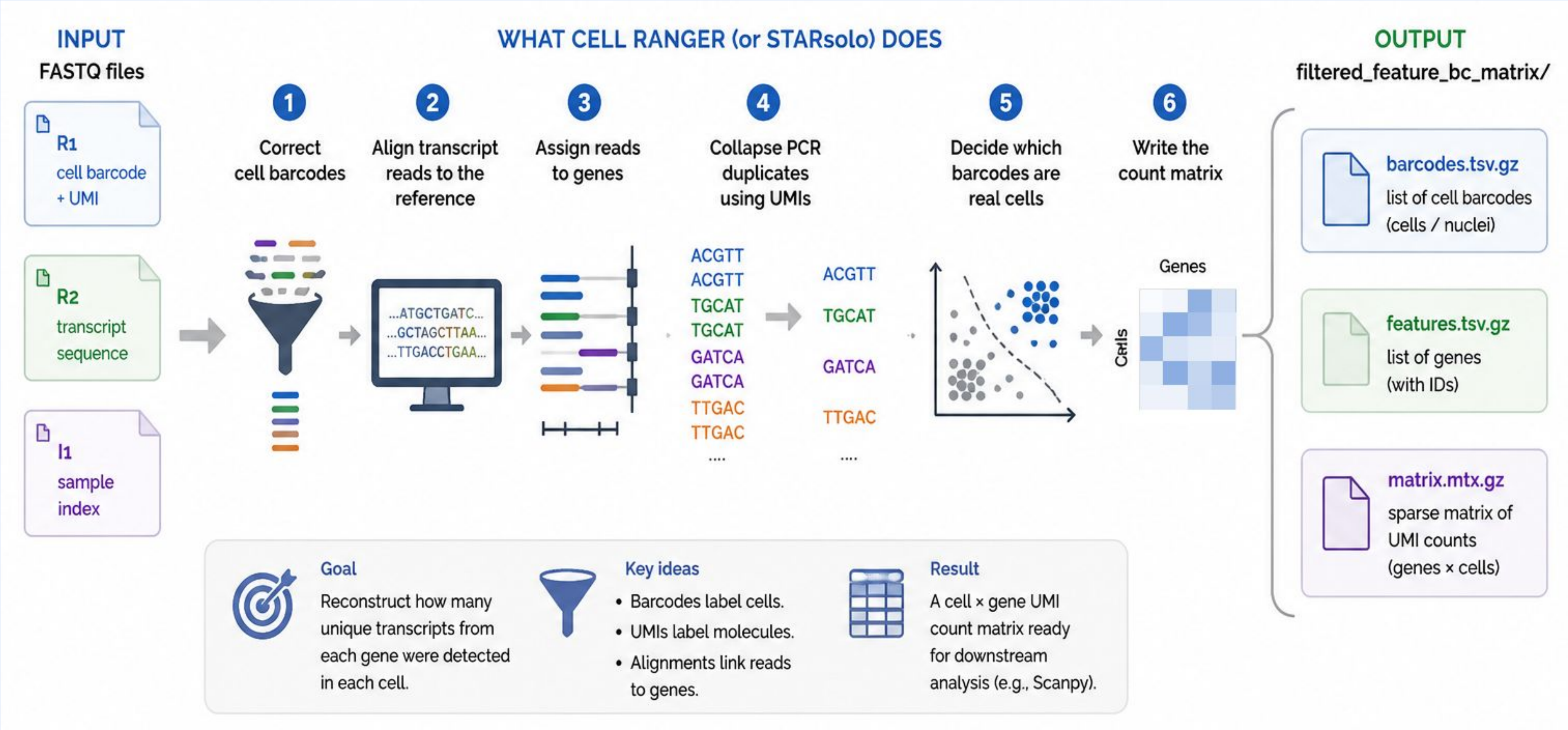
Task 8
clustering similar cell communities
Louvain model
Leiden model

Task 9
discovering what cell-type is each cell
scipy

Task 10
cell-type specific differential expression
DESeq2
(via pydeseq2)

Cell Ranger (or other aligners like STARsolo)

turning reads into a count matrix



R1 tells us: which cell? which molecule?
R2 tells us: which gene?
I1 tells us: which sample/library?

Sources:
10x Genomics Cell Ranger count documentation: https://www.10xgenomics.com/support/software/cell-ranger/latest/analysis/running-pipelines/cr-gex-count
https://www.10xgenomics.com/support/cn/software/cell-ranger/latest/analysis/outputs/cr-outputs-mex-matrices
STARsolo documentation: https://github.com/alexdobin/STAR/blob/master/docs/STARsolo.md

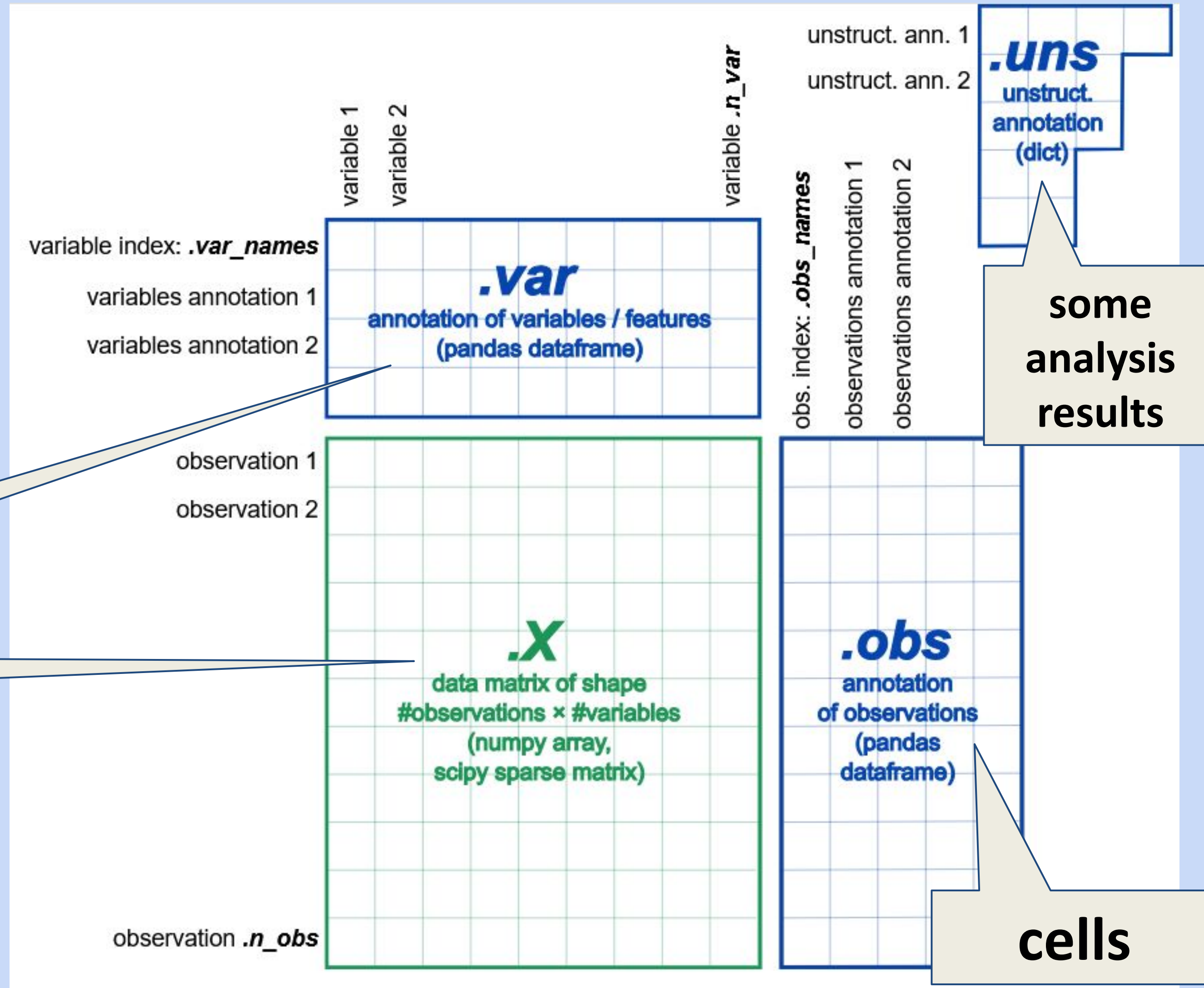
Scanpy AnnData objects

how is the sparse matrix handled in practice?

We will use Scanpy today, but others like **Seurat** too, which stores the data in a different way.

genes

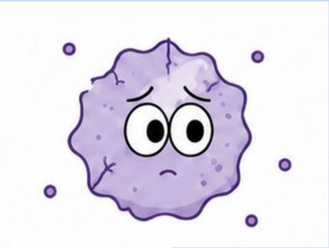
counts

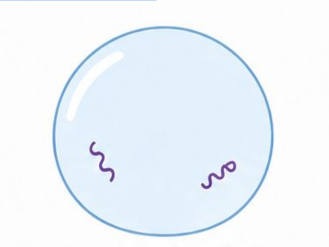


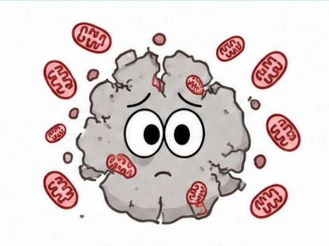
Source: https://scanpy.readthedocs.io/en/stable/usage-principles.html

Quality control

filtering out bad cells



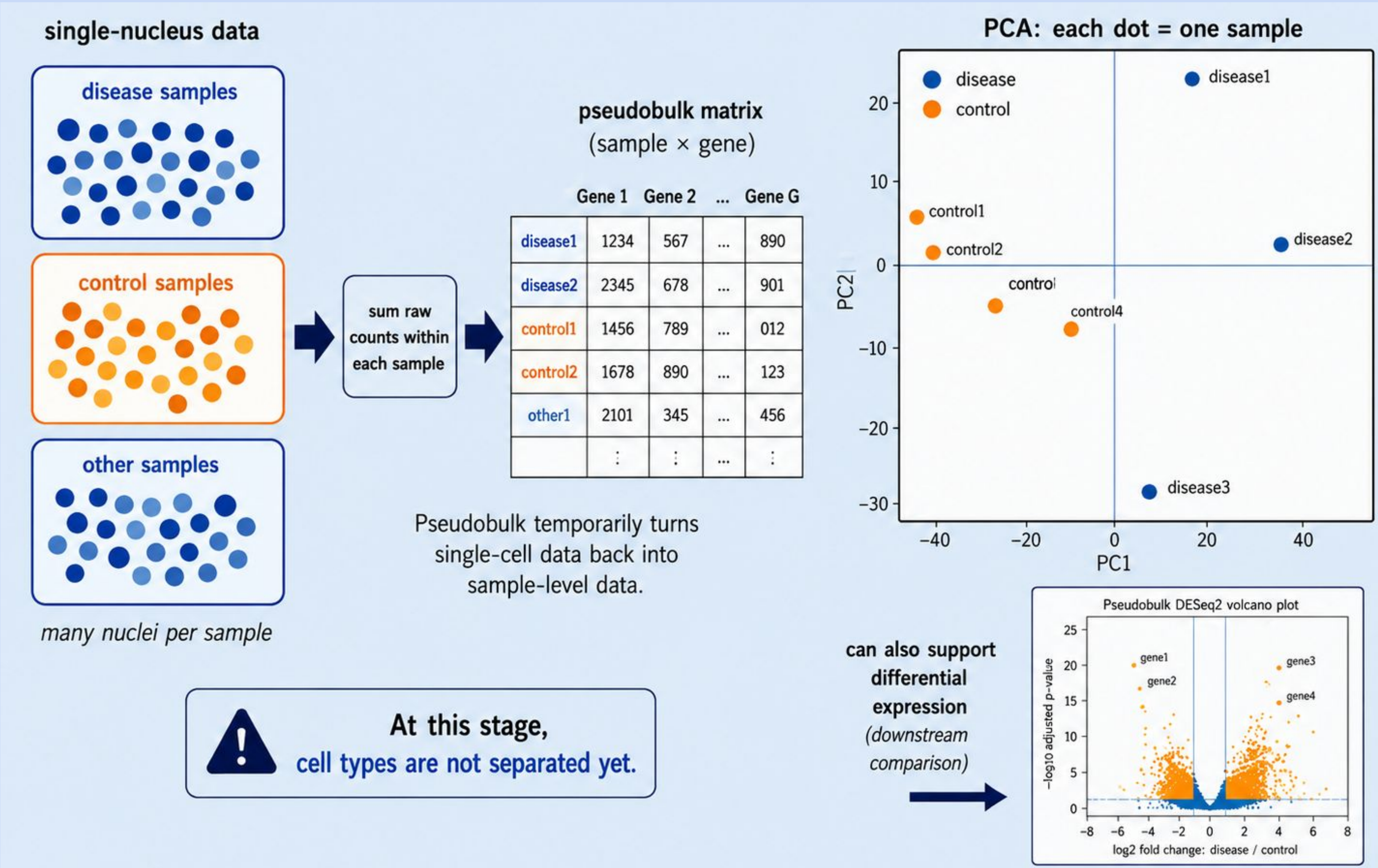




	Usual for single-cell RNA-seq	Usual for single-nucleus RNA-seq	What do we do in this practical?
min_genes	~200-500 genes per cell	~300-500 genes per nucleus	We use **500**. This removes nuclei with too little transcript information.
min_counts	~500-1,000 UMIs per cell	~500-1,000 UMIs per nucleus	We use **500**. In this dataset, the input matrices are already Cell Ranger-filtered, so this is a gentle extra filter.
max_mt	often ~5-20% mitochondrial counts, but tissue-dependent	often lower, ~1-5%, but dataset-dependent	We use **5%**. In nuclei, mitochondrial percentages are usually low; high values catch damaged nuclei / cytoplasmic
min_cells	gene must be detected in ≥3 cells	gene must be detected in ≥3 nuclei	We use **3**. This removes genes seen in too few nuclei to be informative.

We deal with doublets later

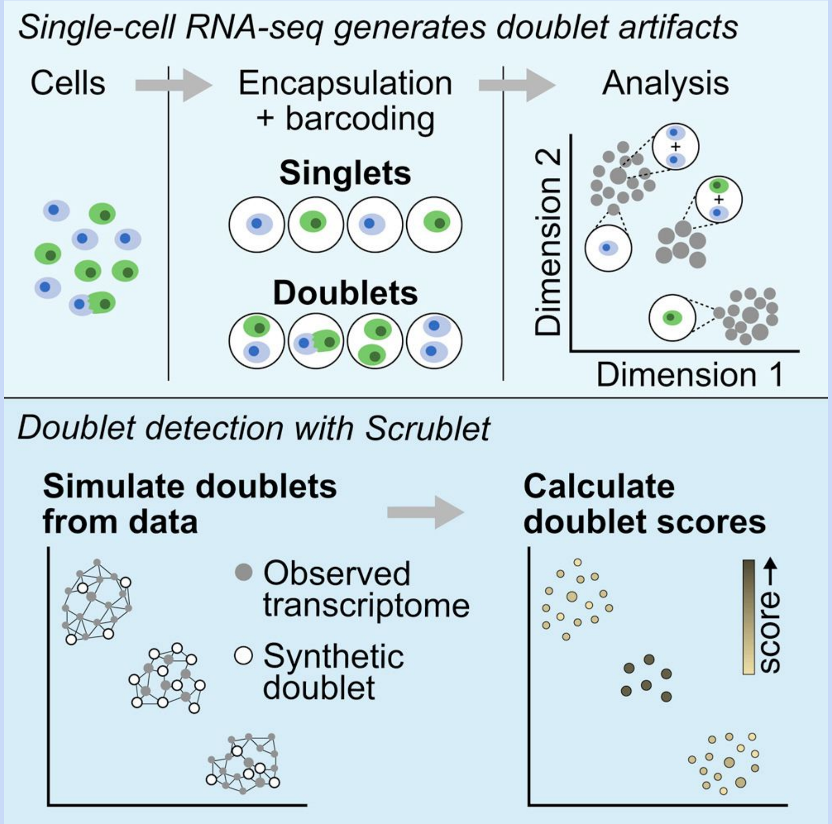
Whole-sample pseudobulk and PCA



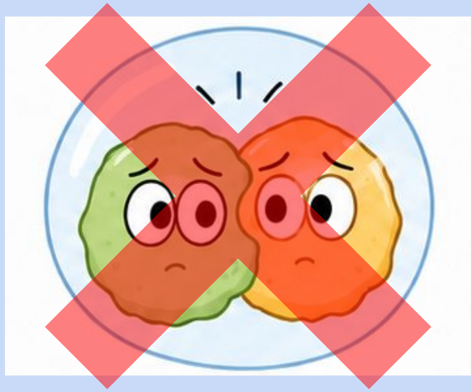
	Gene 1	Gene 2	...	Gene G
disease1	1234	567	...	890
disease2	2345	678	...	901
control1	1456	789	...	012
control2	1678	890	...	123
other1	2101	345	...	456
	⋮	⋮	...	⋮

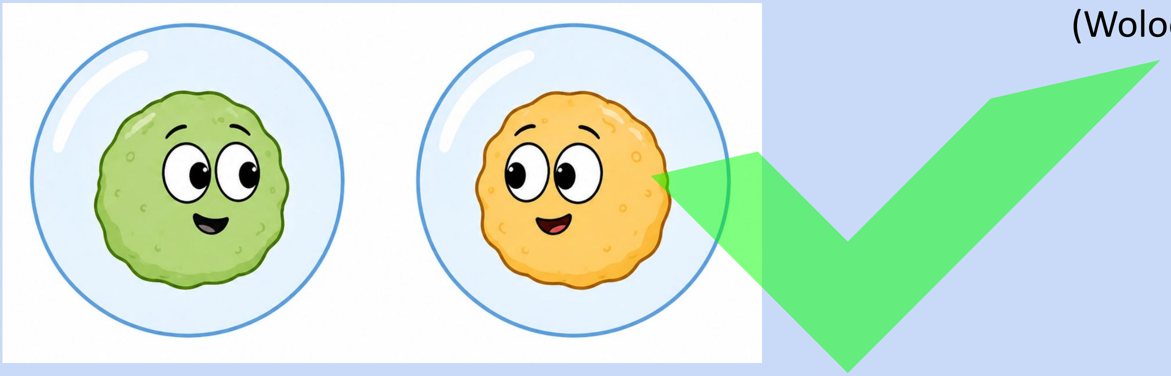
Removing doublets with Scrublet

Step	Scrublet operation	Intuition	Main parameters
1. Simulate doublets	Randomly pairs observed cells and adds their expression profiles to create artificial doublets.	"What would doublets look like in this dataset?"	**R** = number/ratio of simulated doublets
2. Reduce dimensions	Normalizes, selects variable genes, and projects observed + simulated cells into PCA space.	Compare cells in a cleaner low-dimensional space.	**P** = number of PCs
3. Build neighbour graph	Finds nearest neighbours among observed and simulated profiles.	Real cells surrounded by simulated doublets are suspicious.	**k** = number of neighbours
4. Score cells	Calculates a doublet score from how many simulated doublets are nearby.	Higher score = more doublet-like.	**d** = expected doublet rate
5. Call doublets	Applies a threshold to classify likely doublets.	Turns a score into "keep" or "flag/remove."	**τ** = doublet-score threshold



(Wolock 2019)

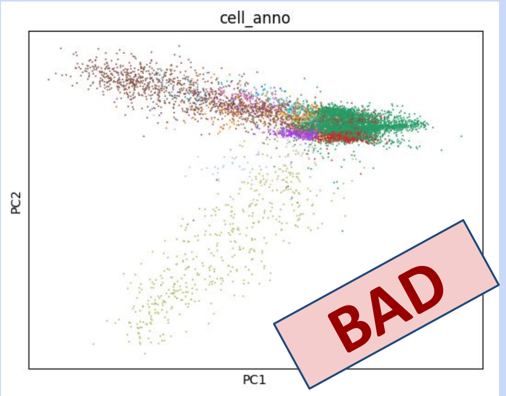




Dimensional reduction

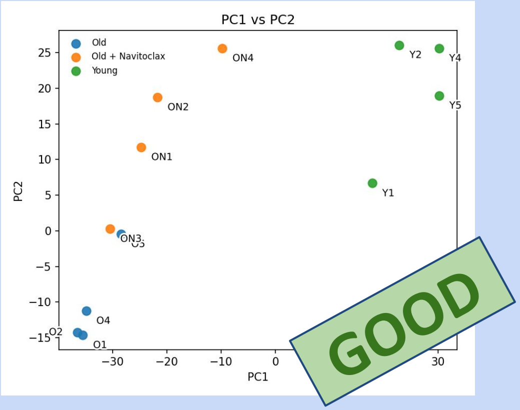
detecting cell clusters

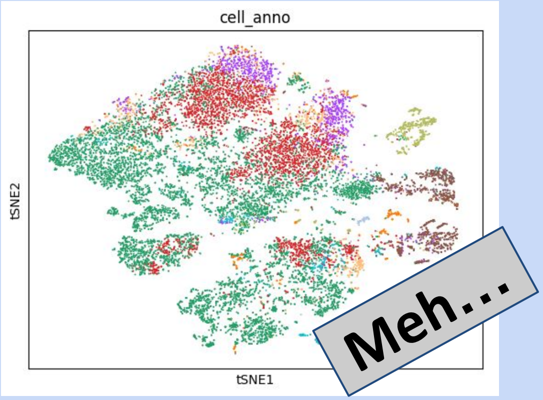
whole-sample analysis



PCA

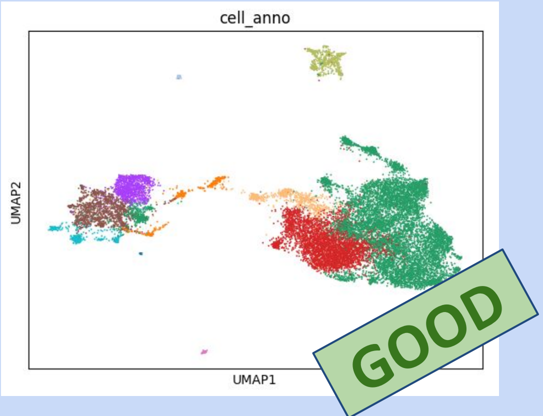
Good for linear dimensional reduction, summarising major sources of variation, and as intermediate step for UMAP later.
Bad to capture biological effects (since they aren't linear).





t-SNE

Good for detecting local neighbourhoods, and non-linear effects. Was more important in the past, now replaced by UMAP.
Bad for small sample numbers.

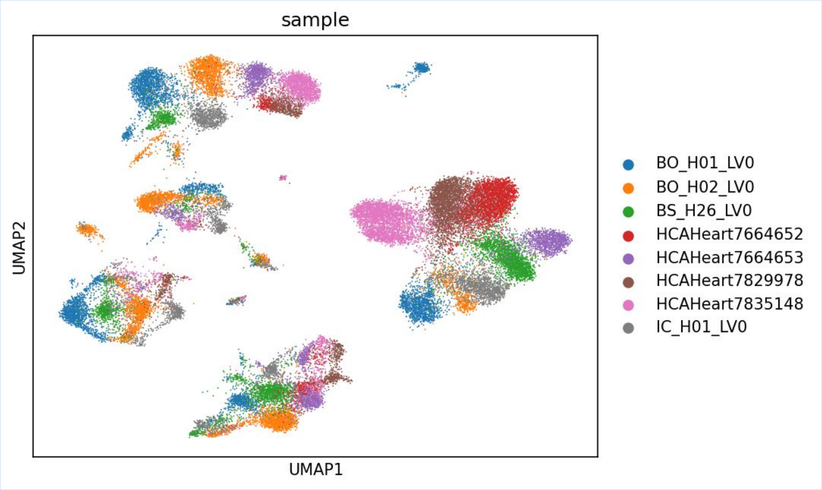


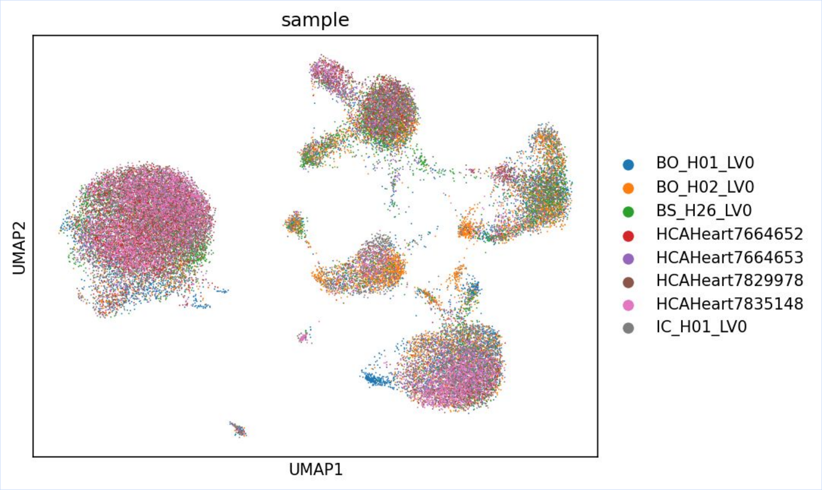
UMAP

Very good for detecting local neighbourhoods, and non-linear effects.
Bad for small sample numbers, and for interpreting the axes of variation.

Batch-effect correction with Harmony

Step	Harmony operation	Intuition	Main parameters
1. Start from PCA + batch labels	Uses PCA coordinates and metadata such as batch, sample, donor, or technology.	Harmony corrects the **embedding**, not the raw count matrix.	Number of PCs chosen upstream
2. Soft-cluster cells	Assigns cells probabilistically to clusters in PCA space.	Similar cells are grouped, but without forcing hard boundaries too early.	**K** = number of clusters **σ** = softness of cluster assignment
3. Encourage batch diversity	Penalizes clusters dominated by one batch.	Similar biological populations should ideally contain cells from multiple batches.	**θ** = diversity penalty higher **θ** forces stronger batch mixing
4. Estimate and subtract batch offsets	Learns cluster-specific batch-effect corrections and subtracts them from PCA coordinates.	Cells are moved to reduce technical separation while preserving biological structure.	**λ** = ridge penalty higher **λ** gives more conservative correction
5. Iterate until convergence	Repeats clustering and correction until the embedding stabilizes.	The final output is a Harmony-corrected PCA embedding for neighbours, UMAP, and clustering.	**ε** = convergence tolerance **T** = maximum iterations





Clustering

Step / concept	Louvain	Leiden
Publication year	Blondel 2008, «Fast unfolding of communities in large networks»	Traag 2019, «From Louvain to Leiden»
Input	A graph: in scRNA-seq, usually a k-nearest-neighbour graph of cells/nuclei.	Same as Louvain.
Goal	Find communities of nodes that are more connected to each other than to the rest of the graph.	Same as Louvain.
Optimisation target	Optimises a graph community score, often modularity.	Same idea, often modularity or CPM; better optimisation procedure.
Local moving step	Moves individual nodes between communities when this improves the score.	Same as Louvain, but uses a faster/local-moving strategy.
Community aggregation	Collapses communities into "super-nodes" and repeats the optimisation on the smaller graph.	Same as Louvain.
Refinement step	No explicit refinement step.	Adds a refinement step before aggregation, improving community quality.
Connectivity guarantee	Can produce communities that are poorly connected or even disconnected.	Guarantees communities are connected, and generally better behaved.
Practical use in scRNA-seq	Older standard; still useful and widely understood.	Usually preferred now; often gives more stable and better-connected clusters.
Main caveat	Clusters are graph communities, not automatically biological cell types.	Same as Louvain.

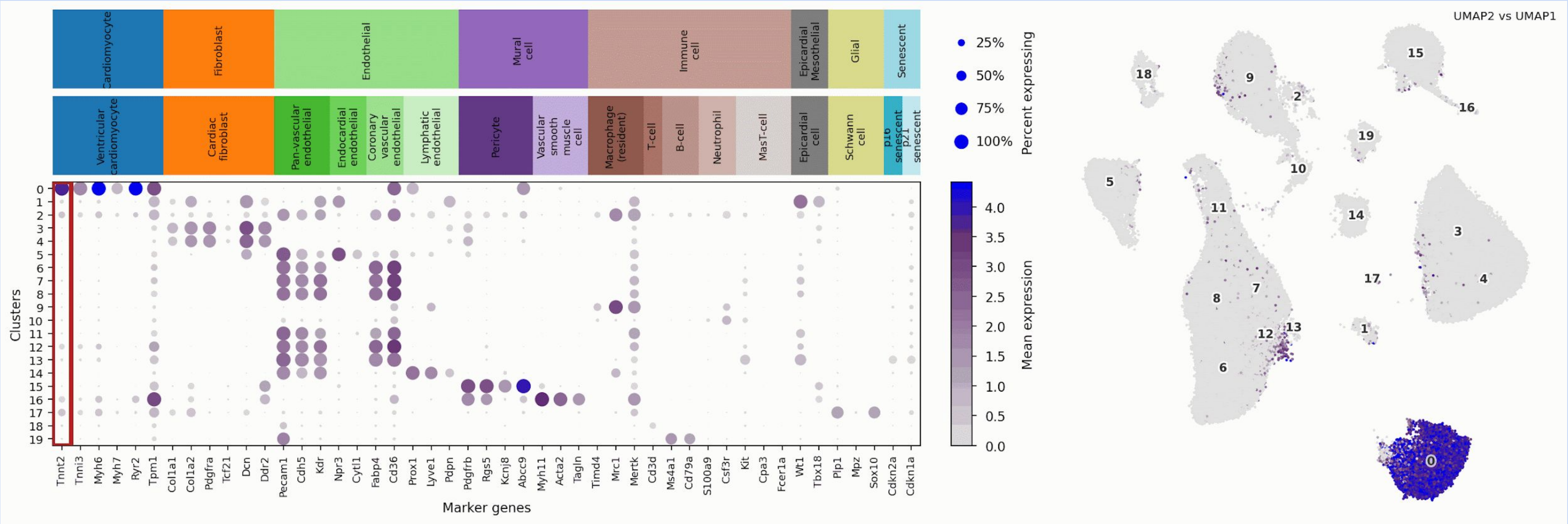
The main algorithm for this operation is Leiden (which replaced the older Louvain).

Clustering uses the nearest-neighbour graph, not visual boundaries drawn on UMAP.

Louvain and Leiden define communities; Leiden is generally better connected.

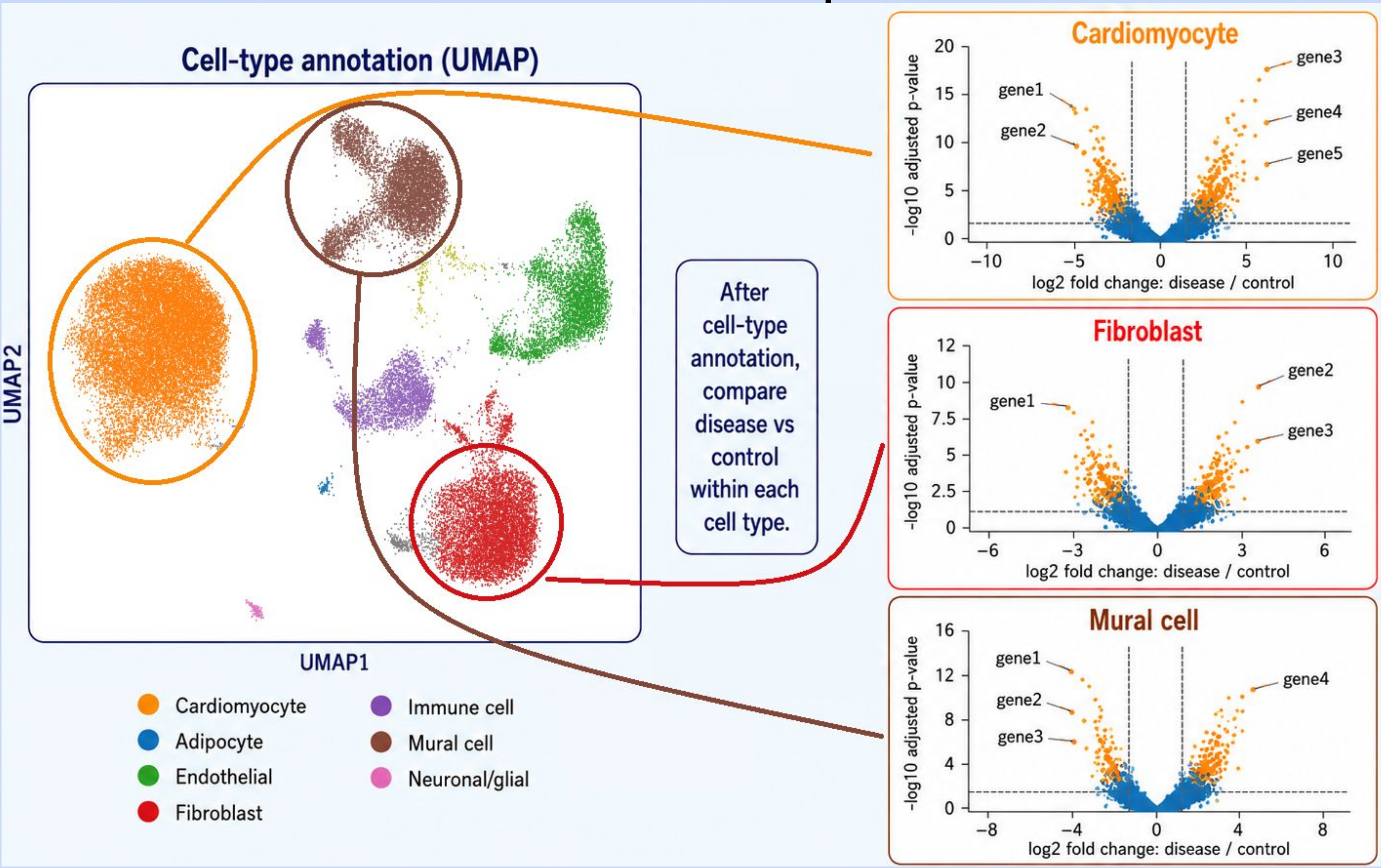
Clusters are hypotheses; marker genes decide whether they match biology.

Cell-type annotation



- Annotation is subjective: combine marker genes, UMAPs, dot plots, and judgement.
- Use several markers together; single markers can mislead in single cell RNA-seq.
- Labels must be evidence-based, revisable, and allowed to remain ambiguous.

Cell-type specific pseudobulk
and differential expression



What next?

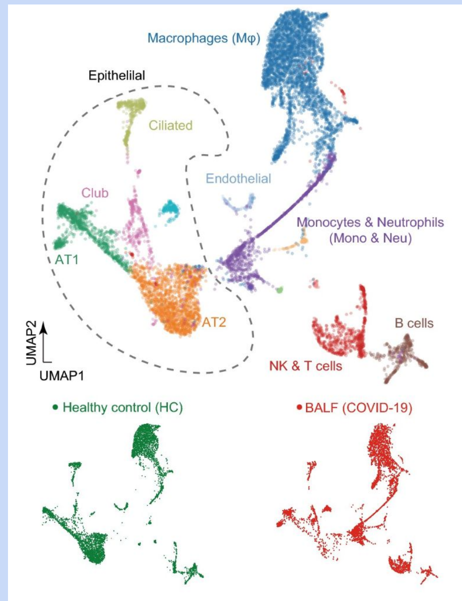
once cells are labelled with their cell-type, we can do many interesting analyses

Compare conditions
• Disease effects within each cell type
• Shared versus cell-specific responses
• Cell-type abundance shifts

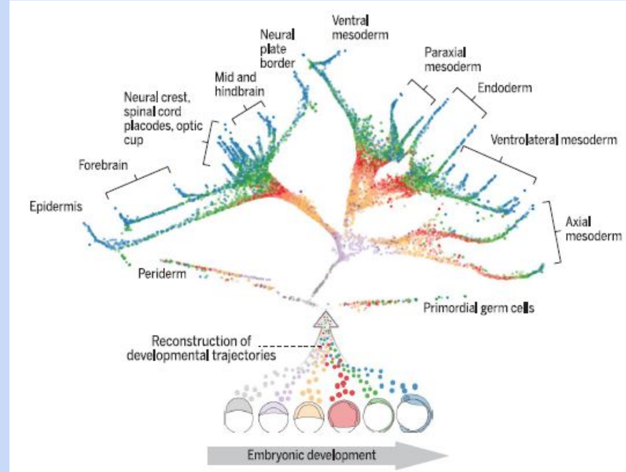
Explore mechanisms
• Ligand-receptor communication hypotheses
• Pathways and gene-set enrichment
• Disease genes in their cellular context

Refine cell identities
• Re-cluster broad cell populations
• Discover finer cell states
• Find better marker genes

Extend the experiment
• Trajectory analysis for dynamic processes
• Spatial transcriptomics for tissue context
• Multi-omics with ATAC-seq or proteins

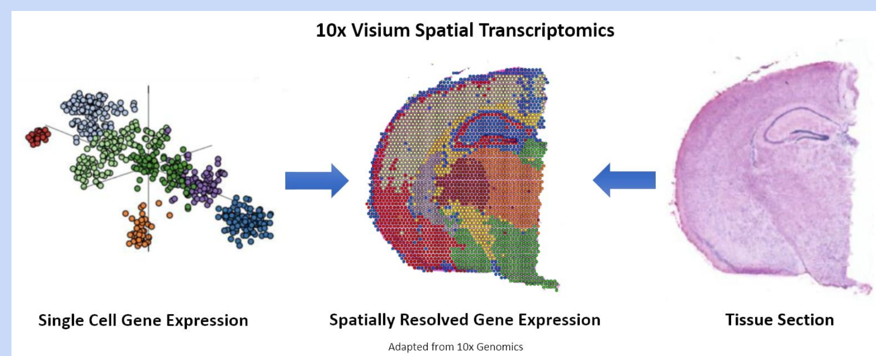


(Jiangping He 2020)



Developmental tree of early zebrafish embryogenesis. Single-cell transcriptomes were generated from zebrafish embryos at 12 developmental stages (six of which are shown). The transcriptional trajectories that describe the fate specification of 25 cell types were reconstructed from the data. Molecular specification is visualized with a force-directed layout, in which each cell is represented by a point (colored by developmental stage), proceeding from pluripotent cells (at the bottom center) outward to 25 distinct cell types. A subset of the identified trajectories are labeled in groups.

(Farrell 2018)



Source: https://sites.dartmouth.edu/cqb/2020/11/24/new-services-sample-multiplexing-spatial-transcriptomics-and-multiomics/

Take-home messages

1. Single-cell avoids averaging the whole sample.

2. A count matrix is not yet biology.

3. Cell identity unlocks many interesting questions.

Let’s play with it!

I propose ten tasks (with fancy names!)

If time is short, we can skip some of the red ones.

Part I: Single cell transcriptomics quality control

1. **Conflux**: importing and merging 10x matrices
2. **Gatekeeper**: quality control and filtering
3. **Census**: checking cell demographics after filtering
4. **Bulklet**: creating pseudobulk RNA-seq summaries

Part II: Embedding and clustering single nuclei

5. **DoubletWatch**: identifying and removing doublets
6. **Maploom**: building the UMAP embedding
7. **Harmonise**: correcting strong batch-effects in our datasets
8. **Clusterforge**: comparing Louvain and Leiden clusters

Part III: Cell-type annotation and differential expression

9. **Cellscribe**: marker genes and cell-type labels
10. **Cellcontrast**: DESeq2 within selected cell types